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Exact Output Steering with Vanishing Controls in Weighted Shift Dynamics

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Abstract

We investigate exact projected steering with asymptotically vanishing initial controls for finite-dimensional wrapped weighted shifts and their residualizations. For wrapped shifts, steering is characterized by an expansive cycle multiplier and a residue separating the source and control channels. For residual shifts, a distinct spectral–toral mechanism emerges: steering through selected conjugate planes is equivalent to radial expansion and an arithmetic parity condition on phase resonances, yielding bounded-gap steering times and quantitative control estimates. As an application, we derive a sufficient two-scale gain condition for steering orbit-dictionary coefficients while preserving their decoded realization across internal depths.

Keywords: Weighted shifts, projected controllability, vanishing controls, linear dynamics, spectral–toral recurrence, orbit-based learning.

1. Introduction

Weighted backward shifts are among the canonical model operators in linear dynamics. Beginning with the classical hypercyclic examples of Rolewicz, they have played a central role in the study of orbit density, recurrence, mixing, and other dynamical properties of linear operators (Bayart and Matheron, 2009). Their significance lies in the combination of simple algebraic structure and rich iterative behavior: explicit formulas for their powers permit precise dynamical classifications that are generally difficult for arbitrary operators. However, the classical phenomena of hypercyclicity and topological mixing rely essentially on infinite-dimensional settings. No linear operator on a finite-dimensional normed space can be hypercyclic.

This raises a natural question: which forms of dynamical reachability remain available for finite-dimensional weighted shifts? We investigate this question through projected steering. Instead of requiring one uncontrolled orbit to visit arbitrary states, we prescribe source and target outputs, allow a small perturbation of the initial state through a fixed control channel, and ask whether the target can be reached exactly after autonomous evolution. The central issue is whether the required perturbation can vanish as the steering horizon increases. This formulation connects the orbit geometry of linear dynamics with one-shot output controllability, while addressing a fundamentally different problem from classical finite-horizon reachability with distributed inputs. Indeed, classical state-to-output and output-to-output controllability have been studied through rank conditions and controllability Gramians (Sarachik and Kreindler, 1965; Danhane et al., 2023), while null controllability with vanishing control energy has been investigated for time-distributed inputs (Pandolfi et al., 2013). Here, we instead restrict control to a single vanishing perturbation

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of the initial state, require exact output transitions along a common sequence of horizons tending to infinity, and characterize the resulting steering property through the cyclic and spectral–toral structure of weighted shifts.

Let $T \in \mathcal{L}(Z)$ act on a finite-dimensional real or complex normed space Z, and let Y be an output space. We consider source and control embeddings $J_{\mathrm{out}}, J_{\mathrm{ctl}} : Y \to Z$ and an output map $P : Z \to Y$ satisfying

$$PJ_{\mathrm{out}} = I_Y, \qquad PJ_{\mathrm{ctl}} = 0.$$

Given arbitrary $u, v \in Y$, we seek controls $h_k(u, v) \to 0$ and a common increasing sequence $n_k \to \infty$ such that

$$PT^{n_k}\big(J_{\mathrm{out}}u + J_{\mathrm{ctl}}h_k(u, v)\big) = v.$$

We require linear dependence of the controls on the source–target pair, with the corresponding operator norms converging to zero. Thus exact output transitions must be achievable uniformly on bounded source–target sets while the initial intervention becomes negligible.

For a wrapped weighted shift, we obtain a complete characterization for arbitrary fixed channels satisfying the stated geometry. Steering is equivalent to cycle expansion, annihilation of the source block at one residue, and invertibility of the corresponding control block. An explicit channel construction attains the output dimension allowed by the source–control geometry.

For real residual shifts with negative cycle multiplier, we use the classical rotation–dilation decomposition and explicitly constructed conjugate-plane channels. Steering for these channels is equivalent to radial expansion and simultaneous angular recurrence. Kronecker's theorem (Cassels, 1957) converts the recurrence requirement into an exact arithmetic condition on the resonance lattice. The resulting control formulas give amplitude bounds and bounded-gap admissible horizons for every fixed positive control budget on bounded source–target sets.

The resulting theory also applies to finite coefficient-coded representations generated by orbit dictionaries. Such representations arise in recent orbit-based learning and structure-guided embedding compression frameworks (Puig, 2026; Puig and Jaiswal, 2026). Applying the steering criterion to a direct-sum lift, we derive a sufficient expansion–contraction balance for preserving decoded realizations across internal orbit depths.

Definition 1 (The wrapped weighted backward shift) *Let $Z = \mathbb{F}^d$, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, with canonical basis $e_0, \dots, e_{d-1}$. A wrapped weighted backward shift $W : Z \to Z$ is defined as $We_j = w_j e_{j-1 \pmod d}$, with $w_j \neq 0$.*

Unlike the standard finite-dimensional backward shift, which is nilpotent, the wrapped shift allows indefinite iteration without collapsing to the zero operator, thereby preserving nontrivial dynamical behavior. The nonzero-weight assumption makes W invertible. Moreover, coordinates circulate modulo the ambient dimension instead of being discarded, and complete cycles produce scalar rescaling. Define the full-cycle multiplier $\Omega := \prod_{j=0}^{d-1} w_j$. Since every coordinate crosses every weight exactly once during d iterations, $W^d = \Omega I$. Consequently, for $n = qd + r$, with $0 \leq r < d$, $W^n = \Omega^q W^r$.

2. Projected Steering Geometry

Output and control geometry. Let Z be a finite-dimensional normed space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $T \in \mathcal{L}(Z)$, and set $Y = \mathbb{F}^p$. Fix injective linear maps $J_{\text{out}}, J_{\text{ctl}} : Y \to Z$ and an output map $P : Z \to Y$ satisfying

$$PJ_{\text{out}} = I_Y, \qquad PJ_{\text{ctl}} = 0.$$

Thus $J_{\text{out}}u$ represents the initial source, while $J_{\text{ctl}}h$ is a perturbation restricted to a fixed subspace of $\ker P$. For each $n \geq 0$, define the source-to-output and control-to-output operators

$$A_n := PT^n J_{\text{out}}, \qquad G_n := PT^n J_{\text{ctl}}.$$

Exact steering from $u \in Y$ to $v \in Y$ at depth n amounts to solving

$$A_n u + G_n h = v.$$

Whenever G_n is invertible, the unique control is $h = G_n^{-1}(v - A_n u)$. The question is whether such controls can be chosen to vanish along a common sequence of increasing depths, uniformly for bounded source–target pairs.

Definition 2 (Steering Criterion) *We say that T satisfies the Steering Criterion along a strictly increasing sequence (n_k) if there exist linear maps $Q_k, S_k : Y \to Y$ such that $\|Q_k\| \to 0$, $\|S_k\| \to 0$, and*

$$PT^{n_k}(J_{\text{out}} + J_{\text{ctl}}Q_k) = 0, \qquad PT^{n_k}J_{\text{ctl}}S_k = I_Y.$$

Our steering criterion says, schematically,

$$\forall u, v \quad \exists (x_k(u,v))_k : \quad x_k(u,v) \to J_{\text{out}}u, \quad PT^{n_k}x_k(u,v) = v,$$

where $x_k(u,v) = J_{\text{out}}u + J_{\text{ctl}}(Q_k u + S_k v)$.

Theorem 3 (Exact block characterization) *Let (n_k) be strictly increasing. The following are equivalent.*

(i) T satisfies the Steering Criterion along (n_k);

(ii) For every k, G_{n_k} is invertible and $\|G_{n_k}^{-1}\| \to 0$, $\|G_{n_k}^{-1}A_{n_k}\| \to 0$.

Proof If (ii) holds, take $Q_k = -G_{n_k}^{-1}A_{n_k}$, $S_k = G_{n_k}^{-1}$. Then $A_{n_k} + G_{n_k}Q_k = 0$ and $G_{n_k}S_k = I_Y$. Conversely, if (i) holds, then $G_{n_k}S_k = I_Y$. Because Y is finite-dimensional and G_{n_k} is a square operator, G_{n_k} is surjective and hence invertible. Therefore $S_k = G_{n_k}^{-1}$. Similarly, $A_{n_k} + G_{n_k}Q_k = 0$ implies $Q_k = -G_{n_k}^{-1}A_{n_k}$. The required limits are exactly $\|S_k\| \to 0$ and $\|Q_k\| \to 0$. ∎

Why projected rather than full-state steering? The Steering Criterion guarantees exact arrival at the prescribed output, $PT^{n_k}x_k(u,v) = v$, whereas full-state asymptotic arrival would require $T^{n_k}x_k(u,v) = J_{\text{out}}v$. We show in Appendix B that, for both expansive steering mechanisms classified below, every $u \neq 0$ and every $v \in Y$ satisfy

$$\|T^{n_k}x_k(u,v) - J_{\text{out}}v\| \longrightarrow \infty.$$

In particular, neither steering mechanism can be asymptotically full-state accurate.

3. Cycle-Synchronous Steering dynamics for W

Let W be a wrapped weighted backward shift, with full-cycle multiplier Ω as in Definition 1. For the fixed channels, write

$$A_r = PW^r J_{\mathrm{out}}, \qquad G_r = PW^r J_{\mathrm{ctl}}, \qquad 0 \le r < d.$$

The cycle identity gives

$$A_{qd+r} = \Omega^q A_r, \qquad G_{qd+r} = \Omega^q G_r.$$

Whenever G_r is invertible,

$$G_{qd+r}^{-1} = \Omega^{-q} G_r^{-1}, \qquad G_{qd+r}^{-1} A_{qd+r} = G_r^{-1} A_r.$$

In particular, the normalized source block is independent of q. For $R > 0$, let $B_Y(0,R) := \{y \in Y : \|y\| < R\}$.

Theorem 4 (Equivalent characterizations of steering for W) *Let W be a wrapped weighted backward shift on $\mathbb{F}^d$, and let $\Omega := \prod_{j=0}^{d-1} w_j$. The following statements are equivalent.*

(i) *W satisfies the Steering Criterion along some strictly increasing sequence.*

(ii) *One has $|\Omega| > 1$, and there exists $r \in \{0, \dots, d-1\}$ such that $A_r = 0$ and G_r is invertible.*

(iii) *There exists $r \in \{0, \dots, d-1\}$ such that $A_r = 0$ and, for every $\varepsilon > 0$ and every $R > 0$, there exists an integer $q_0 \ge 0$ such that for every $q \ge q_0$,*

$$B_Y(0,R) \subseteq PW^{qd+r} J_{\mathrm{ctl}}\big(B_Y(0,\varepsilon)\big).$$

Proof Suppose (i) holds. By Theorem 3, along an increasing sequence (n_k), the blocks G_{n_k} are eventually invertible and

$$\|G_{n_k}^{-1}\| \to 0, \qquad \|G_{n_k}^{-1} A_{n_k}\| \to 0.$$

Passing to an infinite subsequence with a fixed residue r, write $n_k = q_k d + r$, where $q_k \to \infty$. The preceding identities imply that G_r is invertible, $|\Omega| > 1$, and $G_r^{-1} A_r = 0$. Thus $A_r = 0$, proving (ii). Conversely, assume (ii) and take $n_k = kd + r$. Then

$$A_{n_k} = 0, \qquad G_{n_k}^{-1} = \Omega^{-k} G_r^{-1}.$$

Hence the Steering Criterion holds by Theorem 3, with explicit maps

$$Q_k = 0, \qquad S_k = \Omega^{-k} G_r^{-1}.$$

Finally, if (ii) holds, then $PW^{qd+r} J_{\mathrm{ctl}} = \Omega^q G_r$. For every $\varepsilon > 0$,

$$B_Y\left(0, \frac{\varepsilon |\Omega|^q}{\|G_r^{-1}\|}\right) \subseteq PW^{qd+r} J_{\mathrm{ctl}} B_Y(0,\varepsilon).$$

Since $|\Omega| > 1$, this proves (iii). Conversely, (iii) forces G_r to be surjective, hence invertible, because its range contains a nonempty open ball. If $|\Omega| \le 1$, then

$$PW^{qd+r}J_{\mathrm{ctl}}B_Y(0,1) \subseteq B_Y(0, \|G_r\|)$$

for every q, contradicting (iii) for sufficiently large target balls. Thus (ii) follows. ∎

Lemma 5 (Maximal steerable output dimension for W) *If $Z = \mathbb{F}^d$, $Y = \mathbb{F}^p$ and W satisfies the Steering Criterion along some sequence, then $p \le \lfloor \frac{d}{2} \rfloor$.*

Proof By Theorem 4 there exists $0 \le r \le d-1$ such that $A_r = PW^rJ_{\mathrm{out}} = 0$, hence $W^rJ_{\mathrm{out}}Y \subseteq \ker P$. But W^r and J_{out} are injective, so $\dim W^rJ_{\mathrm{out}}Y = p$. Since $\dim \ker P = d - p$, we obtain $p \le d - p$. Therefore $p \le \lfloor \frac{d}{2} \rfloor$. ∎

Canonical steering channels for the wrapped shift. The output and control channels can be constructed explicitly.

Proposition 6 (Canonical steering geometry) *Let $W : \mathbb{F}^d \to \mathbb{F}^d$ be a wrapped weighted backward shift with nonzero weights, and assume $d \ge 2p$. Let $f_0, \dots, f_{p-1}$ be the canonical basis of $Y = \mathbb{F}^p$, and let $e_0, \dots, e_{d-1}$ be the canonical basis of Z. Define $J_{\mathrm{out}}f_j := e_j$, $J_{\mathrm{ctl}}f_j := e_{p+j}$, and*

$$Pe_j = \begin{cases} f_j, & 0 \le j \le p-1, \\ 0, & p \le j \le d-1. \end{cases}$$

Then $PJ_{\mathrm{out}} = I_Y$ and $PJ_{\mathrm{ctl}} = 0$. Moreover, for the residue $r = p$, $A_p = PW^pJ_{\mathrm{out}} = 0$ and $G_p = PW^pJ_{\mathrm{ctl}}$ is invertible.

Proof For $0 \le j < p$, the vector $W^pJ_{\mathrm{out}}f_j$ is supported in the last p coordinates, so $PW^pJ_{\mathrm{out}} = 0$. On the other hand, $W^pJ_{\mathrm{ctl}}f_j$ is a nonzero multiple of e_j. Thus $G_p = PW^pJ_{\mathrm{ctl}}$ is diagonal with nonzero diagonal entries, hence invertible. ∎

Consequently, when $|\Omega| > 1$, every output dimension $1 \le p \le \lfloor d/2 \rfloor$ is attained by channels satisfying the Steering Criterion.

4. Spectral–Toral Steering for $R = I + W$

Let $W : \mathbb{R}^d \to \mathbb{R}^d$ be a real wrapped weighted backward shift with $\Omega = -r^d$, $r > 0$, and set $R = I + W$. Write

$$M = \left\lfloor \frac{d}{2} \right\rfloor, \qquad \delta = d - 2M \in \{0, 1\}.$$

When d is even, $\delta = 0$ and the whole real space decomposes into M invariant two-dimensional conjugate-eigenvalue planes. When d is odd, $\delta = 1$ and there are M such planes together with one additional real eigendirection.

The characteristic polynomial of W is $z^d + r^d$. Its eigenvalues are

$$\nu_j = r \exp\left(\frac{(2j+1)\pi i}{d}\right), \qquad 0 \le j < d.$$

For $0 \le j < M$, define

$$\theta_j = \frac{(2j+1)\pi}{d}, \qquad \lambda_j = 1 + re^{i\theta_j} = \rho_j e^{i\phi_j},$$

where $\rho_j = \sqrt{1 + r^2 + 2r\cos\theta_j}$. The nonreal eigenvalues occur in conjugate pairs, and there is a real isomorphism $\mathcal{U}$ such that

$$\mathcal{U}^{-1} R^n \mathcal{U} = \left(\bigoplus_{j=0}^{M-1} \rho_j^n \operatorname{Rot}(-n\phi_j) \right) \oplus [(1-r)^n]^\delta.$$

The characteristic-polynomial calculation, the similarity, and the construction of $\mathcal{U}$ are given in Appendix A.

Choose a nonempty set of conjugate-plane indices

$$J = \{j_1, \dots, j_p\} \subseteq \{0, \dots, M-1\}.$$

Write

$$\widehat{Z} = \left(\bigoplus_{j=0}^{M-1} \mathbb{R}^2 \right) \oplus \mathbb{R}^\delta.$$

For each selected plane j_ℓ, let $\widehat{J}_{\text{out}} f_\ell$ and $\widehat{J}_{\text{ctl}} f_\ell$ be the vectors supported on that plane with respective coordinates $(1,0)$ and $(0,1)$. Define

$$\widehat{P}((a_j, b_j)_{j=0}^{M-1}, t) = (a_{j_1}, \dots, a_{j_p}),$$

where t is omitted when d is even. Transporting these maps through the spectral similarity U, set

$$J_{\text{out}} = \mathcal{U}\widehat{J}_{\text{out}}, \qquad J_{\text{ctl}} = \mathcal{U}\widehat{J}_{\text{ctl}}, \qquad P = \widehat{P}\mathcal{U}^{-1}.$$

Then

$$PJ_{\text{out}} = I_Y, \qquad PJ_{\text{ctl}} = 0, \qquad Y = \mathbb{R}^p.$$

All details in Appendix A.3. Equip $Y = \mathbb{R}^p$ with its Euclidean norm. Now, direct calculation (see Lemma 13) gives

$$A_n = \operatorname{diag}_{1\le\ell\le p}\big(\rho_{j_\ell}^n \cos(n\phi_{j_\ell})\big), \quad \text{and} \quad G_n = \operatorname{diag}_{1\le\ell\le p}\big(\rho_{j_\ell}^n \sin(n\phi_{j_\ell})\big).$$

Whenever G_n is invertible,

$$\|G_n^{-1} A_n\| = \max_{1\le\ell\le p} |\cot(n\phi_{j_\ell})|, \quad \text{and} \quad \|G_n^{-1}\| = \max_{1\le\ell\le p} \rho_{j_\ell}^{-n} |\csc(n\phi_{j_\ell})|. \tag{1}$$

Thus the Steering Criterion separates into simultaneous angular recurrence and radial expansion. Set

$$\alpha_J = \left(\frac{\phi_{j_1}}{\pi}, \dots, \frac{\phi_{j_p}}{\pi} \right) \in \mathbb{T}^p, \qquad \beta = \left(\frac{1}{2}, \dots, \frac{1}{2} \right),$$

and define $H_{\alpha_J} = \overline{\{n\alpha_J : n \in \mathbb{Z}\}} \subseteq \mathbb{T}^p$. These identities lead to the following exact classification. Remember that a set $\mathcal{N} \subseteq \mathbb{N}$ is called syndetic if there exists an integer $L \ge 1$ such that for every $N \ge 1$, every interval $\{N, N+1, \dots, N+L\} \subset \mathbb{N}$, contains at least one element of $\mathcal{N}$.

Theorem 7 (Exact spectral–toral classification for R) *Let $R = I + W$ on $Z = \mathbb{R}^d$, where W is a real wrapped weighted backward shift with nonzero weights and full-cycle multiplier $\Omega = -r^d$, $r > 0$. Let $M = \lfloor \frac{d}{2} \rfloor$, choose $J = \{j_1, \dots, j_p\} \subseteq \{0, \dots, M-1\}$, and define the channels P, J_{out}, and J_{ctl} through the selected conjugate planes as above.*

For every $\ell = 1, \dots, p$, define $\theta_{j_\ell} = \frac{(2j_\ell+1)\pi}{d}$, $\lambda_{j_\ell} = 1 + re^{i\theta_{j_\ell}} = \rho_{j_\ell} e^{i\phi_{j_\ell}}$, and $\alpha_J = \left(\frac{\phi_{j_1}}{\pi}, \dots, \frac{\phi_{j_p}}{\pi}\right) \in \mathbb{T}^p$. Then the following are equivalent:

(i) *R satisfies the Steering Criterion along some strictly increasing sequence $n_k \longrightarrow \infty$ with respect to the selected conjugate-plane channels.*

(ii) *The selected radial factors $\rho_{j_\ell} > 1$ $\forall \ell = 1, \dots, p$, and $\beta = \left(\frac{1}{2}, \dots, \frac{1}{2}\right) \in H_{\alpha_J}$.*

(iii) *The selected radial factors $\rho_{j_\ell} > 1$ $\forall \ell = 1, \dots, p$, and the parity-resonance condition holds: $q \cdot \alpha_J \in \mathbb{Z} \implies \sum_{\ell=1}^p q_\ell$ is even, $\forall q \in \mathbb{Z}^p$.*

(iv) *The selected radial factors $\rho_{j_\ell} > 1$ $\forall \ell = 1, \dots, p$, and, for every $\varepsilon > 0$, the phase-recurrence set $\mathcal{N}_\varepsilon := \{n \geq 1 : \|n\alpha_J - \beta\|_{\mathbb{T}^p} < \varepsilon\}$ is syndetic in $\mathbb{N}$.*

Proof sketch. By Theorem 3, steering along (n_k) is equivalent to

$$\|G_{n_k}^{-1}\| \to 0, \qquad \|G_{n_k}^{-1} A_{n_k}\| \to 0.$$

Equation 1 shows that the second limit holds precisely when $\cot(n_k \phi_{j_\ell}) \to 0$ for every ℓ, which is equivalent to $n_k \alpha_J \to \beta$ in $\mathbb{T}^p$ by Lemma 14. Such an increasing sequence exists if and only if $\beta \in H_{\alpha_J}$. Along this sequence, $|\sin(n_k \phi_{j_\ell})| \to 1$ for every selected mode. Hence the first block limit holds if and only if $\rho_{j_\ell} > 1$, $1 \leq \ell \leq p$. This proves the equivalence of (i) and (ii). Kronecker's closure theorem (Cassels, 1957), gives

$$H_{\alpha_J} = \{x \in \mathbb{T}^p : q \cdot x \in \mathbb{Z} \text{ whenever } q \cdot \alpha_J \in \mathbb{Z}\}.$$

Since $q \cdot \beta = \frac{1}{2} \sum_{\ell=1}^p q_\ell$, conditions (ii) and (iii) are equivalent. Finally, translation by α_J is minimal on the compact group H_{α_J}. Therefore every nonempty relatively open neighborhood of β has syndetic return times. Conversely, syndetic returns to arbitrarily small neighborhoods imply $\beta \in H_{\alpha_J}$. This proves the equivalence with (iv).

This establishes the equivalence of the four conditions. The complete proof, including the auxiliary spectral and recurrence arguments, is given in Appendix A.5. ■

Expansion and reachability do not suffice. Consider $r = \sqrt{3}$ and

$$W_r = \begin{pmatrix} 0 & r \\ -r & 0 \end{pmatrix}, \qquad R_r = I + W_r,$$

with $J_{\mathrm{out}} u = (u, 0)$, $J_{\mathrm{ctl}} h = (0, h)$, and $P(x_0, x_1) = x_0$. Here $\rho = \sqrt{1 + r^2} = 2 > 1$. Since $P R_r (J_{\mathrm{out}} u + J_{\mathrm{ctl}} h) = u + rh$, every target v is reachable at horizon one by choosing $h = (v - u)/r$. Nevertheless, $\phi = \arctan r = \pi/3$, and whenever G_n is invertible,

$$\|G_n^{-1} A_n\| = |\cot(n\phi)| = \frac{1}{\sqrt{3}}.$$

Thus the source-cancellation controls cannot vanish, and the Steering Criterion fails despite radial expansion and finite-time output reachability. Equivalently, $\alpha = \phi/\pi = 1/3$ violates the parity-resonance condition of Theorem 7, since $3\alpha = 1 \in \mathbb{Z}$ while 3 is odd. This two-dimensional counterexample shows that the arithmetic condition in Theorem 7 captures a genuine obstruction not detected by expansion and reachability alone.

Control budgets and admissible horizons. For any fixed positive control budget and bounded source–target pairs, exact steering is possible at horizons with bounded gaps. Assume the equivalent conditions of Theorem 7, and set $\rho_{\min} = \min_{1\le\ell\le p} \rho_{j_\ell} > 1$. For $0 < \varepsilon < 1/2$ and $n \in \mathcal{N}_\varepsilon$, the unique exact steering control satisfies

$$\|h_n(u,v)\| \le \tan(\pi\varepsilon)\|u\| + \frac{\rho_{\min}^{-n}}{\cos(\pi\varepsilon)}\|v\|.$$

Indeed, phase proximity gives

$$|\cot(n\phi_{j_\ell})| \le \tan(\pi\varepsilon), \qquad |\csc(n\phi_{j_\ell})| \le \frac{1}{\cos(\pi\varepsilon)},$$

and the estimate follows from the diagonal control formulas (7).

Given $B > 0$ and $\eta > 0$, choose ε so that $B\tan(\pi\varepsilon) < \eta/2$. For all sufficiently large $n \in \mathcal{N}_\varepsilon$, the second term is also below $\eta/2$ whenever $\|v\| \le B$. Thus the horizons admitting exact controls of norm at most η uniformly for $\|u\|, \|v\| \le B$ form a syndetic set. No uniform gap bound as $\eta \downarrow 0$ is asserted.

The radial expansivity condition in Theorem 7 admits an explicit threshold when all conjugate rotation–dilation planes are selected. Set

$$\kappa_d = \begin{cases} 1, & d \text{ is even}, \\ 2, & d \text{ is odd}. \end{cases}$$

Then $\min_{0\le j<M} \rho_j^2 = 1 + r^2 - 2r\cos\left(\frac{\kappa_d\pi}{d}\right)$. Consequently, all conjugate residual modes are expansive if and only if $r > 2\cos\left(\frac{\kappa_d\pi}{d}\right)$. The proof is given in Appendix A.6.

5. Application to Orbit-Dictionary Coefficients

Finite orbit dictionaries represent outputs by coefficient vectors. We apply the preceding steering criteria to these coefficients and derive a sufficient gain condition for preserving a decoded realization across internal orbit depths. The construction applies, in particular, to the orbit-based representations considered in (Puig, 2026) and (Puig and Jaiswal, 2026).

Finite orbit representations. Let $X = \mathbb{F}^{d_{\text{scaff}}}$, $V = \mathbb{F}^C$, and let $W_{\text{scaff}} \in \mathcal{L}(X)$ be a wrapped weighted shift. Fix finite blocks $D_1, \dots, D_K$, vectors $x_{k,j} \in X$, $1 \le j \le |D_k|$, and linear maps $J_k : X \to V$. Set

$$N_{\text{sub}} = \sum_{k=1}^{K} |D_k|, \qquad Y_{\text{fr}} = \mathbb{F}^{d_{\text{scaff}} N_{\text{sub}}}.$$

For $c = (c_{k,m,j}) \in Y_{\mathrm{fr}}$, define the depth-dependent synthesis maps

$$\mathscr{D}_s(c) = \sum_{k=1}^{K} \sum_{j=1}^{|D_k|} \sum_{m=0}^{d_{\mathrm{scaff}}-1} c_{k,m,j} J_k W_{\mathrm{scaff}}^{m+s} x_{k,j}.$$

In particular, $\mathscr{D}_0$ is the synthesis map of the fixed orbit dictionary.

Commuting steering dynamics. Let $T_{\mathrm{steer}} \in \mathcal{L}(Z)$ satisfy the Steering Criterion on Y_{fr}. On $\mathcal{Z} = Z \oplus X^{N_{\mathrm{sub}}}$, define

$$\mathbb{T} = T_{\mathrm{steer}} \oplus I, \qquad \mathbb{W} = I_Z \oplus W_{\mathrm{scaff}}^{\oplus N_{\mathrm{sub}}}.$$

These operators commute by construction, since they act on separate summands. Writing $\mathbf{x} = (x_{k,j})_{k,j}$, define the decoder

$$\mathfrak{D}(z, \mathbf{x}) = \sum_{k,j,m} (Pz)_{k,m,j} J_k W_{\mathrm{scaff}}^{m} x_{k,j},$$

where the indices range over the finite orbit dictionary. Theorem 4 with Proposition 6, or Theorem 7, supplies explicit steering operators under the corresponding hypotheses. By definition, for every $z \in Z$ and $s \geq 0$,

$$\mathfrak{D}\Big(z, (W_{\mathrm{scaff}}^{\oplus N_{\mathrm{sub}}})^s \mathbf{x}\Big) = \mathscr{D}_s(Pz).$$

Theorem 8 (Controlled framework navigation) *Suppose T_{steer} satisfies the Steering Criterion along $n_q \to \infty$, with steering maps Q_q, S_q. Fix $c_0 \in Y_{\mathrm{fr}}$, and let $(c_q) \subset Y_{\mathrm{fr}}$ satisfy $\|S_q c_q\| \to 0$. For arbitrary internal depths $s_q \geq 0$, set*

$$h_q = Q_q c_0 + S_q c_q, \qquad \xi_q = (J_{\mathrm{out}} c_0 + J_{\mathrm{ctl}} h_q, \mathbf{x}).$$

Then $h_q \to 0$, $\xi_q \to (J_{\mathrm{out}} c_0, \mathbf{x})$, and

$$\mathfrak{D}(\mathbb{T}^{n_q} \mathbb{W}^{s_q} \xi_q) = \mathfrak{D}(\mathbb{W}^{s_q} \mathbb{T}^{n_q} \xi_q) = \mathscr{D}_{s_q}(c_q). \tag{2}$$

Proof Since the Steering Criterion guarantees $\|Q_q\| \to 0$, our assumption $\|S_q c_q\| \to 0$ implies $\|h_q\| \leq \|Q_q\|\|c_0\| + \|S_q c_q\| \longrightarrow 0$. Since J_{ctl} is bounded, $J_{\mathrm{ctl}} h_q \to 0$, and therefore $\xi_q \to (J_{\mathrm{out}} c_0, \mathbf{x})$. The Steering Criterion gives, $PT_{\mathrm{steer}}^{n_q}(J_{\mathrm{out}} c_0 + J_{\mathrm{ctl}} h_q) = PT_{\mathrm{steer}}^{n_q}(J_{\mathrm{out}} + J_{\mathrm{ctl}} Q_q) c_0 + PT_{\mathrm{steer}}^{n_q} J_{\mathrm{ctl}} S_q c_q = 0 + c_q = c_q$. The commutation of the two lifted operators gives

$$\begin{aligned}
\mathfrak{D}\left(\mathbb{T}_{\mathrm{steer}}^{n_q} \mathbb{W}_{\mathrm{scaff}}^{s_q} \xi_q\right) &= \sum_{k=1}^{K} \sum_{m=0}^{d_{\mathrm{scaff}}-1} \sum_{j=1}^{|D_k|} \left[PT_{\mathrm{steer}}^{n_q}(J_{\mathrm{out}} c_0 + J_{\mathrm{ctl}} h_q)\right]_{k,m,j} J_k W_{\mathrm{scaff}}^{m+s_q} x_{k,j} \\
&= \sum_{k=1}^{K} \sum_{m=0}^{d_{\mathrm{scaff}}-1} \sum_{j=1}^{|D_k|} (c_q)_{k,m,j} J_k W_{\mathrm{scaff}}^{m+s_q} x_{k,j} \quad = \mathscr{D}_{s_q}(c_q).
\end{aligned}$$

■

Every bounded target-code sequence satisfies $\|S_q c_q\| \to 0$.

Preserving realizations across depths. Let $\Omega_{\mathrm{scaff}} \neq 0$ be the full-cycle multiplier of W_{scaff}. Define the coefficient shift C_{scaff} by

$$(C_{\mathrm{scaff}}c)_{k,m,j} = \begin{cases} c_{k,m+1,j}, & 0 \le m < d_{\mathrm{scaff}} - 1, \\ \Omega_{\mathrm{scaff}}^{-1} c_{k,0,j}, & m = d_{\mathrm{scaff}} - 1. \end{cases}$$

The wrapped-cycle identity implies

$$\mathscr{D}_s(C_{\mathrm{scaff}}^s c) = \mathscr{D}_0(c), \qquad C_{\mathrm{scaff}}^{d_{\mathrm{scaff}}} = \Omega_{\mathrm{scaff}}^{-1} I. \tag{3}$$

Proof of (3) in Appendix C. Identities in (3) show that internal orbit evolution can be compensated exactly, preserving the same depth-zero realization across arbitrary scaffold depths. However, when $0 < |\Omega_{\mathrm{scaff}}| < 1$, this preservation requires coefficient amplification as the depth increases, potentially preventing the steering controls from vanishing. The following result establishes an explicit tradeoff between internal scaffold depth and external steering horizon, guaranteeing exact preservation of the realization with asymptotically vanishing initial controls.

Corollary 9 (Two-scale gain condition) *In the setting of Theorem 8, fix $c_* \in Y_{\mathrm{fr}}$, and let $s_q = a_q d_{\mathrm{scaff}} + t_q$, $0 \le t_q < d_{\mathrm{scaff}}$. Choose $c_0 = c_*$ and $c_q = C_{\mathrm{scaff}}^{s_q} c_*$. Consequently, the fixed realization $\mathscr{D}_0(c_*)$ is preserved exactly with vanishing initial controls whenever*

$$\|S_q\| |\Omega_{\mathrm{scaff}}|^{-a_q} \longrightarrow 0. \tag{4}$$

For cycle-synchronous steering with $n_q = q\, d_{\mathrm{steer}} + r$, it is sufficient that $|\Omega_{\mathrm{steer}}|^q |\Omega_{\mathrm{scaff}}|^{a_q} \longrightarrow \infty$, whereas spectral–toral steering admits the sufficient condition $\rho_{\min}^{n_q} |\Omega_{\mathrm{scaff}}|^{a_q} \longrightarrow \infty$ along the recurrent steering horizons.

Proof By the contragredient identities (3),

$$\mathscr{D}_{s_q}(c_q) = \mathscr{D}_0(c_*), \qquad c_q = \Omega_{\mathrm{scaff}}^{-a_q} C_{\mathrm{scaff}}^{t_q} c_*.$$

Hence

$$\|S_q c_q\| \le M_{\mathrm{scaff}} \|S_q\| |\Omega_{\mathrm{scaff}}|^{-a_q} \|c_*\|,$$

where $M_{\mathrm{scaff}} = \max_{0 \le t < d_{\mathrm{scaff}}} \|C_{\mathrm{scaff}}^t\|$. Theorem 8 then applies whenever (4) is satisfied, yielding

$$\mathfrak{D}\left(\mathbb{T}_{\mathrm{steer}}^{n_q} \mathbb{W}_{\mathrm{scaff}}^{s_q} \xi_q\right) = \mathscr{D}_{s_q}(c_q) = \mathscr{D}_0(c_*).$$

To conclude, remember that for cycle-synchronous steering $\|S_q\| = |\Omega_{\mathrm{steer}}|^{-q} \|G_r^{-1}\|$, while for the spectral–toral steering case, for all sufficiently large q, $\|S_q\| \le 2\rho_{\min}^{-n_q}$. ∎

Extending this construction to redundant orbit charts, including OrBIT, requires compatibility with chart gluing and is left for future work.

Appendix A. Spectral-Toral Steering Dynamics for $R = I + W$

The following elementary spectral calculations and classical recurrence facts are included to make the channel construction and Theorem 7 self-contained.

Lemma 10 (Characteristic polynomial and eigenvalues of W) *The minimal and characteristic polynomials of W are both equal to $\mu_W(z) = \chi_W(z) = z^d + r^d$. Consequently, the eigenvalues of W are simple and equal to*

$$\nu_\ell = r \exp\left(\frac{(2\ell+1)\pi i}{d}\right), \qquad \ell = 0, \dots, d-1.$$

Proof Since $W^d = -r^d I_{\mathbb{R}^d}$, the minimal polynomial μ_W divides $z^d + r^d$. On the other hand, the vectors

$$e_0, We_0, \dots, W^{d-1}e_0$$

are nonzero multiples of distinct canonical basis vectors, since each iteration shifts the coordinate index by one modulo d and all weights are nonzero. They are therefore linearly independent, so no nonzero polynomial of degree less than d annihilates W. Thus $\deg \mu_W = d$, and hence

$$\mu_W(z) = z^d + r^d.$$

Since the characteristic polynomial χ_W is monic of degree d and divisible by μ_W, we obtain $\chi_W = \mu_W$. The eigenvalues are consequently the d distinct roots of $z^d = -r^d$, namely

$$\nu_\ell = r \exp\left(\frac{(2\ell+1)\pi i}{d}\right), \qquad \ell = 0, \dots, d-1.$$

■

Since $R = I + W$, the eigenvalues of R are $\lambda_\ell = 1 + \nu_\ell$. For $0 \le j < M$, recall that

$$\theta_j = \frac{(2j+1)\pi}{d}, \qquad \lambda_j = 1 + re^{i\theta_j} = \rho_j e^{i\phi_j}.$$

Since $0 < \theta_j < \pi$, we have $\operatorname{Im} \lambda_j = r \sin\theta_j > 0$, so $\phi_j \in (0, \pi)$. Moreover,

$$\rho_j^2 = |1 + re^{i\theta_j}|^2 = 1 + r^2 + 2r\cos\theta_j.$$

Thus $\rho_j > 1$ if and only if $r + 2\cos\theta_j > 0$. The remaining nonreal eigenvalues occur in conjugate pairs, since $\nu_{d-1-j} = \overline{\nu_j}$. When d is odd, the additional real eigenvalue of W is $\nu_M = -r$, yielding the real eigenvalue $1 - r$ of R.

A.1. Diagonal similarity to a canonical wrapped shift

The spectrum above can also be obtained through an explicit diagonal similarity. This similarity shows that the spectrum of a wrapped weighted backward shift depends on the weights only through their full-cycle product.

Define the canonical wrapped weighted backward shift $C : \mathbb{R}^d \to \mathbb{R}^d$ by $Ce_0 = -r^d e_{d-1}$ and $Ce_j = e_{j-1},\ j = 1, \dots, d-1$. Thus C is the wrapped weighted backward shift whose weights are $(-r^d, 1, \dots, 1)$. Define nonzero scalars $\gamma_0 := 1$ and

$$\gamma_j := \frac{1}{w_1 w_2 \cdots w_j}, \qquad j = 1, \dots, d-1.$$

Let $D : \mathbb{R}^d \to \mathbb{R}^d$ be the invertible diagonal operator defined by $De_j = \gamma_j e_j$.

Lemma 11 (Explicit diagonal similarity) *One has $D^{-1}WD = C$. Equivalently, $W = DCD^{-1}$.*

Proof For $1 \le j < d$, the definition of γ_j gives

$$D^{-1}WDe_j = \frac{\gamma_j w_j}{\gamma_{j-1}} e_{j-1} = e_{j-1} = Ce_j.$$

For $j = 0$, using $\Omega = \prod_{j=0}^{d-1} w_j = -r^d$,

$$D^{-1}WDe_0 = \frac{w_0}{\gamma_{d-1}} e_{d-1} = \Omega e_{d-1} = Ce_0.$$

Thus $D^{-1}WD = C$. ∎

The canonical shift C has an explicit eigenvector associated with every root ν of $\nu^d = -r^d$. Indeed, define $v(\nu) := (1, \nu, \nu^2, \dots, \nu^{d-1})^\top \in \mathbb{C}^d$. For every root ν of $\nu^d = -r^d$,

$$Cv(\nu) = (\nu, \nu^2, \dots, \nu^d)^\top = \nu v(\nu).$$

Since $W = DCD^{-1}$, it follows that $z(\nu) := Dv(\nu)$ satisfies

$$Wz(\nu) = WDv(\nu) = DCv(\nu) = \nu z(\nu).$$

A.2. Real rotation–dilation decomposition

We next pass from the complex eigenvalue description to a real block decomposition. For every $j = 0, \dots, M-1$, let $z_j := z(\nu_j) \in \mathbb{C}^d$ be a complex eigenvector satisfying $Wz_j = \nu_j z_j$. Write $z_j = a_j + ib_j,\ a_j, b_j \in \mathbb{R}^d$. The vectors a_j and b_j are linearly independent over $\mathbb{R}$. Indeed, suppose that they were linearly dependent. Then z_j would be a nonzero complex scalar multiple of a real vector $a \in \mathbb{R}^d$. Thus $z_j = ca$ for some $c \in \mathbb{C}\setminus\{0\}$. Since $Wz_j = \nu_j z_j$, we would obtain $cWa = \nu_j ca$, and therefore $Wa = \nu_j a$. The left-hand side is real, whereas ν_j is nonreal and $a \neq 0$. This is impossible. Hence $\dim_{\mathbb{R}} \operatorname{span}_{\mathbb{R}}\{a_j, b_j\} = 2$. Define the real plane $E_j := \operatorname{span}_{\mathbb{R}}\{a_j, b_j\}$. Because $Rz_j = \lambda_j z_j$ and $\lambda_j = \rho_j(\cos\phi_j + i\sin\phi_j)$, we have $R(a_j + ib_j) = \rho_j(\cos\phi_j + i\sin\phi_j)(a_j + ib_j)$. Expanding the right-hand side gives

$$R(a_j + ib_j) = \rho_j[\cos\phi_j\, a_j - \sin\phi_j\, b_j] + i\rho_j[\sin\phi_j\, a_j + \cos\phi_j\, b_j].$$

Comparing real and imaginary parts, we obtain $Ra_j = \rho_j(\cos\phi_j\, a_j - \sin\phi_j\, b_j)$ and $Rb_j = \rho_j(\sin\phi_j\, a_j + \cos\phi_j\, b_j)$.

If d is odd, let $z_* := z(-r) \in \mathbb{R}^d$. Then $Wz_* = -rz_*$ and $Rz_* = (1-r)z_*$. Define $E_* := \operatorname{span}_{\mathbb{R}}\{z_*\}$.

Because the eigenvalues of W are all distinct, the corresponding complex eigenvectors form a basis of $\mathbb{C}^d$. Grouping conjugate pairs and, when d is odd, the remaining real eigenvector, we obtain the real direct-sum decomposition

$$\mathbb{R}^d = \bigoplus_{j=0}^{M-1} E_j \oplus E_*^{\delta}.$$

Here the last summand is omitted when $\delta = 0$. Define the real model space

$$\widehat{Z} := \left(\bigoplus_{j=0}^{M-1} \mathbb{R}^2 \right) \oplus \mathbb{R}^{\delta}.$$

For

$$\xi = ((x_0, y_0), \dots, (x_{M-1}, y_{M-1}), t) \in \widehat{Z},$$

where the coordinate t is omitted when $\delta = 0$, define

$$\mathcal{U}\xi := \sum_{j=0}^{M-1} (x_j a_j + y_j b_j) + \delta\, t z_*. \tag{5}$$

The preceding direct-sum decomposition shows that $\mathcal{U} : \widehat{Z} \to \mathbb{R}^d$ is a real linear isomorphism.

Proposition 12 (Real rotation–dilation similarity) *With the notation above,*

$$\mathcal{U}^{-1} R^n \mathcal{U} = \left(\bigoplus_{j=0}^{M-1} \rho_j^n \operatorname{Rot}(-n\phi_j) \right) \oplus [1-r]^{n\delta}.$$

Proof On the j-th copy of $\mathbb{R}^2$, $\mathcal{U}_j(x,y) = xa_j + yb_j$. Then, $R\mathcal{U}_j(x,y) = xRa_j + yRb_j = \rho_j(x\cos\phi_j + y\sin\phi_j)\, a_j + \rho_j(-x\sin\phi_j + y\cos\phi_j)\, b_j$. Applying $\mathcal{U}_j^{-1}$ returns the coordinates

$$\mathcal{U}_j^{-1} R \mathcal{U}_j(x,y) = \rho_j \begin{pmatrix} \cos\phi_j & \sin\phi_j \\ -\sin\phi_j & \cos\phi_j \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \rho_j \operatorname{Rot}(-\phi_j)(x,y).$$

Thus, on the j-th two-dimensional summand, $\mathcal{U}_j^{-1} R \mathcal{U}_j = \rho_j \operatorname{Rot}(-\phi_j)$. If d is odd, $\mathcal{U}$ maps the canonical basis of the final one-dimensional summand to z_*, and $Rz_* = (1-r)z_*$. Hence the final one-dimensional block is $[1-r]$. Combining all invariant summands yields $\mathcal{U}^{-1} R \mathcal{U} = \left(\bigoplus_{j=0}^{M-1} \rho_j \operatorname{Rot}(-\phi_j) \right) \oplus [1-r]^{\delta}$.

Moreover, $(\mathcal{U}^{-1} R \mathcal{U})^2 = \mathcal{U}^{-1} R \mathcal{U} \mathcal{U}^{-1} R \mathcal{U} = \mathcal{U}^{-1} R^2 \mathcal{U}$. So, for every $n \geq 1, (\mathcal{U}^{-1} R \mathcal{U})^n = \mathcal{U}^{-1} R^n \mathcal{U}$. Therefore, the corresponding diagram commutes:

$$\begin{array}{ccc} \mathbb{R}^d & \xrightarrow{\quad R^n \quad} & \mathbb{R}^d \\ \mathcal{U}^{-1} \downarrow & & \downarrow \mathcal{U}^{-1} \\ \widehat{Z} & \xrightarrow{\left(\bigoplus_{j=0}^{M-1} \rho_j^n \operatorname{Rot}(-n\phi_j)\right) \oplus [1-r]^{n\delta}} & \widehat{Z}. \end{array}$$

To justify the lower horizontal arrow, observe that $\mathrm{Rot}(\phi)\,\mathrm{Rot}(\psi) = \mathrm{Rot}(\phi+\psi)$. Hence, by induction, $\mathrm{Rot}(\phi)^n = \mathrm{Rot}(n\phi)$. Therefore $(\rho_j\,\mathrm{Rot}(-\phi_j))^n = \rho_j^n\,\mathrm{Rot}(-n\phi_j)$. ∎

The purpose of $\mathcal{U}$ is to provide a fixed real coordinate system in which the conjugate spectral modes become explicit rotation–dilation planes. We will use these coordinates to define the source, control, and output channels on the original space.

A.3. Source and control blocks.

Choose a set of conjugate-plane indices $J \subseteq \{0,\dots,M-1\}$. Write $J = \{j_1,\dots,j_p\}$, where $p := |J| \le \lfloor \frac{d}{2} \rfloor$. Let $Y := \mathbb{R}^p$ with canonical basis $f_1,\dots,f_p$. For every selected plane E_{j_ℓ}, we use: the first coordinate of the plane as the embedded source coordinate; the second coordinate of the plane as the control coordinate; and the first coordinate of the plane as the output readout.

We first define the channels on the model space $\widehat{Z}$. Let $\widehat{J}_{\mathrm{out}} : Y \to \widehat{Z}$ be defined by

$$\widehat{J}_{\mathrm{out}} f_\ell = \Bigg(0,\dots,0,\underbrace{(1,0)}_{j_\ell\text{-th plane}},0,\dots,0\Bigg). \tag{6}$$

Likewise, define $\widehat{J}_{\mathrm{ctl}} : Y \to \widehat{Z}$ by

$$\widehat{J}_{\mathrm{ctl}} f_\ell = \Bigg(0,\dots,0,\underbrace{(0,1)}_{j_\ell\text{-th plane}},0,\dots,0\Bigg).$$

Define $\widehat{P} : \widehat{Z} \to Y$ by

$$\widehat{P}\,((x_0,y_0),\dots,(x_{M-1},y_{M-1}),t) = \left(x_{j_1},\dots,x_{j_p}\right),$$

where the final coordinate t is omitted when d is even.

Thus $\widehat{P}$ reads the first coordinate of every selected conjugate plane and ignores: the second coordinate of every selected plane; both coordinates of every unselected plane; and the remaining real eigendirection when d is odd.

The model-space channels satisfy $\widehat{P}\widehat{J}_{\mathrm{out}} = I_Y$ and $\widehat{P}\widehat{J}_{\mathrm{ctl}} = 0$. Indeed, for every canonical basis vector f_ℓ, $\widehat{P}\widehat{J}_{\mathrm{out}} f_\ell = f_\ell$, whereas $\widehat{P}\widehat{J}_{\mathrm{ctl}} f_\ell = 0$.

We now transport these channels to the original state space $Z = \mathbb{R}^d$. Define $J_{\mathrm{out}} := \mathcal{U}\widehat{J}_{\mathrm{out}}$, $J_{\mathrm{ctl}} := \mathcal{U}\widehat{J}_{\mathrm{ctl}}$, and $P := \widehat{P}\mathcal{U}^{-1}$. Then $J_{\mathrm{out}}, J_{\mathrm{ctl}} : Y \to Z$ are injective, while $P : Z \to Y$ is surjective. Moreover,

$$PJ_{\mathrm{out}} = \widehat{P}\mathcal{U}^{-1}\mathcal{U}\widehat{J}_{\mathrm{out}} = \widehat{P}\widehat{J}_{\mathrm{out}} = I_Y,$$

and

$$PJ_{\mathrm{ctl}} = \widehat{P}\mathcal{U}^{-1}\mathcal{U}\widehat{J}_{\mathrm{ctl}} = \widehat{P}\widehat{J}_{\mathrm{ctl}} = 0.$$

Thus these are legitimate output and control channels in the notation of the Steering Criterion. For every $n \ge 0$, define $A_n := PR^nJ_{\mathrm{out}}$ and $G_n := PR^nJ_{\mathrm{ctl}}$.

Lemma 13 (Exact source and control blocks) *For every $n \ge 0$,*

$$A_n = \mathrm{diag}_{1\le\ell\le p}\left(\rho_{j_\ell}^n \cos(n\phi_{j_\ell})\right) \quad \text{and} \quad G_n = \mathrm{diag}_{1\le\ell\le p}\left(\rho_{j_\ell}^n \sin(n\phi_{j_\ell})\right).$$

Proof Fix $\ell \in \{1, \dots, p\}$. By construction, $\widehat{J}_{\text{out}} f_\ell$ is the first canonical coordinate vector $(1, 0)$ in the plane indexed by j_ℓ, and $J_{\text{out}} f_\ell = \mathcal{U} \widehat{J}_{\text{out}} f_\ell$. By Proposition 12, we have $\mathcal{U}^{-1} R^n \mathcal{U} = \left(\bigoplus_{j=0}^{M-1} \rho_j^n \operatorname{Rot}(-n\phi_j) \right) \oplus [1-r]^{n\delta}$, so we obtain $A_n f_\ell = P R^n J_{\text{out}} f_\ell = \widehat{P} \mathcal{U}^{-1} R^n \mathcal{U} \widehat{J}_{\text{out}} f_\ell$. On the selected plane j_ℓ,

$$\rho_{j_\ell}^n \operatorname{Rot}(-n\phi_{j_\ell}) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \rho_{j_\ell}^n \begin{pmatrix} \cos(n\phi_{j_\ell}) \\ -\sin(n\phi_{j_\ell}) \end{pmatrix}.$$

The map $\widehat{P}$ reads the first coordinate of this plane. Therefore $A_n f_\ell = \rho_{j_\ell}^n \cos(n\phi_{j_\ell}) f_\ell$. Since this holds for every ℓ, $A_n = \operatorname{diag}_{1 \le \ell \le p} \left(\rho_{j_\ell}^n \cos(n\phi_{j_\ell}) \right)$.

Similarly, $\widehat{J}_{\text{ctl}} f_\ell$ is the second canonical coordinate vector $(0, 1)$ in the plane indexed by j_ℓ. On that plane,

$$\rho_{j_\ell}^n \operatorname{Rot}(-n\phi_{j_\ell}) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \rho_{j_\ell}^n \begin{pmatrix} \sin(n\phi_{j_\ell}) \\ \cos(n\phi_{j_\ell}) \end{pmatrix}.$$

Reading the first coordinate gives $G_n f_\ell = \rho_{j_\ell}^n \sin(n\phi_{j_\ell}) f_\ell$. Thus, $G_n = \operatorname{diag}_{1 \le \ell \le p} \left(\rho_{j_\ell}^n \sin(n\phi_{j_\ell}) \right)$. ■

The operator G_n is invertible if and only if $\sin(n\phi_{j_\ell}) \neq 0$ for every $\ell = 1, \dots, p$. Whenever this holds, $G_n^{-1} = \operatorname{diag}_{1 \le \ell \le p} \left(\rho_{j_\ell}^{-n} \csc(n\phi_{j_\ell}) \right)$. Furthermore,

$$\begin{aligned} G_n^{-1} A_n &= \operatorname{diag}_{1 \le \ell \le p} \left(\rho_{j_\ell}^{-n} \csc(n\phi_{j_\ell}) \right) \operatorname{diag}_{1 \le \ell \le p} \left(\rho_{j_\ell}^n \cos(n\phi_{j_\ell}) \right) = \operatorname{diag}_{1 \le \ell \le p} \left(\csc(n\phi_{j_\ell}) \cos(n\phi_{j_\ell}) \right) \\ &= \operatorname{diag}_{1 \le \ell \le p} \left(\cot(n\phi_{j_\ell}) \right). \end{aligned}$$

Thus the source-correction maps and target-control maps determined by the exact block characterization are

$$Q_n = -G_n^{-1} A_n = -\operatorname{diag}_{1 \le \ell \le p} \left(\cot(n\phi_{j_\ell}) \right), \quad S_n = G_n^{-1} = \operatorname{diag}_{1 \le \ell \le p} \left(\rho_{j_\ell}^{-n} \csc(n\phi_{j_\ell}) \right). \tag{7}$$

We equip $Y = \mathbb{R}^p$ with its Euclidean norm. Hence the operator norm of a diagonal operator $\operatorname{diag}(c_1, \dots, c_p)$ is $\max_{1 \le \ell \le p} |c_\ell|$. Therefore $\|G_n^{-1} A_n\| = \max_{1 \le \ell \le p} |\cot(n\phi_{j_\ell})|$ and $\|G_n^{-1}\| = \max_{1 \le \ell \le p} \left[\rho_{j_\ell}^{-n} |\csc(n\phi_{j_\ell})| \right]$. The Steering Criterion for these channels is therefore equivalent to finding an increasing sequence (n_k) such that simultaneously $\cot(n_k \phi_{j_\ell}) \longrightarrow 0$ and $\rho_{j_\ell}^{-n_k} |\csc(n_k \phi_{j_\ell})| \longrightarrow 0$ for every $\ell = 1, \dots, p$. The first family of conditions is purely rotational. The second combines rotational recurrence with radial expansion.

A.4. The torus condition

Use α_J, β, and H_{α_J} as defined in Section 4.

Lemma 14 (Phase convergence and cotangent collapse) *Let (t_k) be a sequence of real numbers. Then $\cot(t_k) \longrightarrow 0$ if and only if $\frac{t_k}{\pi} \longrightarrow \frac{1}{2}$ in $\mathbb{T}$. In either case, $|\sin t_k| \longrightarrow 1$ and $|\csc t_k| \longrightarrow 1$.*

Proof The equivalence follows from the continuity and π-periodicity of the cotangent, whose zeros are precisely $\pi/2$ modulo π. The remaining limits follow from $1+\cot^2 t = \csc^2 t$. ■

The following is the standard minimal-rotation property of the compact cyclic group H_{α_J}; see (Eisner et al., 2015, Theorem 10.13).

Lemma 15 (Existence of an increasing phase-recurrence sequence) *There exists a strictly increasing sequence $n_k \longrightarrow \infty$ such that $n_k\alpha_J \longrightarrow \beta$ in $\mathbb{T}^p$ if and only if $\beta \in H_{\alpha_J}$.*

A.5. Detailed proof of Theorem 7 (Exact spectral–toral classification for R)

The equivalence between (ii) and (iii) follows from Kronecker's closure theorem (Cassels, 1957) and the parity-resonance calculation above.

(ii) implies (i). Assume that $\rho_{j_\ell} > 1$ for every $\ell = 1, \dots, p$ and $\beta \in H_{\alpha_J}$. By Lemma 15, there exists a strictly increasing sequence $n_k \to \infty$ such that $n_k\alpha_J \to \beta$ in $\mathbb{T}^p$. Lemma 14 then gives

$$\max_{1\le\ell\le p} |\cot(n_k\phi_{j_\ell})| \longrightarrow 0, \qquad |\csc(n_k\phi_{j_\ell})| \longrightarrow 1 \quad \text{for every } \ell.$$

In particular, G_{n_k} is eventually invertible. By Lemma 13,

$$\|G_{n_k}^{-1}A_{n_k}\| = \max_{1\le\ell\le p} |\cot(n_k\phi_{j_\ell})| \longrightarrow 0.$$

Since all selected radial factors exceed one,

$$\|G_{n_k}^{-1}\| = \max_{1\le\ell\le p} \rho_{j_\ell}^{-n_k} |\csc(n_k\phi_{j_\ell})| \longrightarrow 0.$$

Theorem 3 therefore gives (i).

(i) implies (ii). Assume that the Steering Criterion holds along a strictly increasing sequence $n_k \to \infty$. By Theorem 3, $\|G_{n_k}^{-1}\| \longrightarrow 0$, and $\|G_{n_k}^{-1}A_{n_k}\| \longrightarrow 0$. Lemma 13 shows that $\cot(n_k\phi_{j_\ell}) \to 0$ for every ℓ. By Lemma 14, $n_k\alpha_J \to \beta$ in $\mathbb{T}^p$, so $\beta \in H_{\alpha_J}$, and $|\csc(n_k\phi_{j_\ell})| \longrightarrow 1$ for every ℓ. Since

$$\rho_{j_\ell}^{-n_k} |\csc(n_k\phi_{j_\ell})| \le \|G_{n_k}^{-1}\| \longrightarrow 0,$$

we must have $\rho_{j_\ell} > 1$ for every selected mode. Thus (ii) holds.

(ii) *implies* (iv). Assume that $\rho_{j_\ell} > 1$ $\forall \ell = 1, \dots, p$, and that $\beta \in H_{\alpha_J}$. Fix $\varepsilon > 0$ and define the relative neighborhood

$$U_\varepsilon := \{x \in H_{\alpha_J} : \|x - \beta\|_{\mathbb{T}^p} < \varepsilon\}.$$

Since $\beta \in H_{\alpha_J}$, the set U_ε is a nonempty relatively open subset of H_{α_J}. Consider the translation

$$\tau_{\alpha_J} : H_{\alpha_J} \to H_{\alpha_J}, \qquad \tau_{\alpha_J}(x) := x + \alpha_J.$$

Since H_{α_J} is the compact cyclic group generated by α_J, the translation τ_{α_J} is minimal (Eisner et al., 2015, Theorem 10.13). By the classical uniform-recurrence theorem (Furstenberg, 1981), the return-time set

$$N_\varepsilon = \{n \ge 1 : \tau_{\alpha_J}^n(0) \in U_\varepsilon\}$$

is syndetic. Thus (iv) holds.

(iv) *implies* (ii). Assume condition (iv). The radial condition $\rho_{j_\ell} > 1\ \forall \ell = 1, \dots, p$ is already part of (iv). It remains to prove that $\beta \in H_{\alpha_J}$. For every integer $k \geq 1$, set $\varepsilon_k := \frac{1}{k}$. Suppose we have already chosen $n_{k-1} \in \mathcal{N}_{\varepsilon_{k-1}}$. Since $\mathcal{N}_{\varepsilon_k}$ is syndetic, hence infinite, we can choose $n_k \in \mathcal{N}_{\varepsilon_k}$ such that $n_{k-1} < n_k$. Then we have an increasing sequence (n_k) such that $\|n_k \alpha_J - \beta\|_{\mathbb{T}^p} < \frac{1}{k}$, for all k. Consequently, $n_k \alpha_J \longrightarrow \beta$ in $\mathbb{T}^p$. By Lemma 15, $\beta \in H_{\alpha_J}$. Thus condition (ii) holds. ■

A.6. Smallest residual radial factor for expansivity

Lemma 16 (Smallest residual radial factor) *The sequence $(\rho_j)_{j=0}^{M-1}$ is strictly decreasing. Consequently, $\min_{0 \leq j \leq M-1} \rho_j = \rho_{M-1}$. Moreover, $\theta_{M-1} = \pi - \frac{\kappa_d \pi}{d}$, and hence $\rho_{M-1}^2 = 1 + r^2 - 2r \cos\left(\frac{\kappa_d \pi}{d}\right)$. Therefore all conjugate residual modes are expansive if and only if $r > 2 \cos\left(\frac{\kappa_d \pi}{d}\right)$.*

Proof Since $0 < \theta_0 < \cdots < \theta_{M-1} < \pi$ and cosine is strictly decreasing on $(0, \pi)$, the identity $\rho_j^2 = 1 + r^2 + 2r \cos \theta_j$ shows that $(\rho_j)_{j=0}^{M-1}$ is strictly decreasing. Hence $\min_{0 \leq j < M} \rho_j = \rho_{M-1}$. Since $d = 2M + \delta$, with $\delta \in \{0, 1\}$, and $\kappa_d = 1 + \delta$, we have

$$\theta_{M-1} = \frac{(2M-1)\pi}{d} = \pi - \frac{\kappa_d \pi}{d}.$$

Consequently,

$$\rho_{M-1}^2 = 1 + r^2 - 2r \cos\left(\frac{\kappa_d \pi}{d}\right).$$

All conjugate residual modes are expansive if and only if $\rho_{M-1} > 1$, equivalently, since $r > 0$, to $r > 2 \cos\left(\frac{\kappa_d \pi}{d}\right)$. ■

Appendix B. Obstruction to Asymptotic Full-State Steering

Proposition 17 (Obstruction to asymptotic full-state steering) *Let $T \in \mathcal{L}(Z)$ be in either of the following settings:*
(i) *$T = W$ is a wrapped weighted backward shift satisfying $|\Omega| > 1$.*
(ii) *$T = R = I + W$, with selected conjugate planes $J = \{j_1, \dots, j_p\}$ satisfying $\rho_{j_\ell} > 1$ for every ℓ, and with the corresponding channels J_{out} and J_{ctl}.*

Let (n_k) be any strictly increasing sequence and let $Q_k, S_k : Y \to Y$ satisfy $\|Q_k\| \to 0$ and $\|S_k\| \to 0$. For $u, v \in Y$, define $x_k(u, v) := J_{\text{out}} u + J_{\text{ctl}}(Q_k u + S_k v)$.

Then, for every $u \neq 0$ and every $v \in Y$,

$$\|T^{n_k} x_k(u, v) - J_{\text{out}} v\| \longrightarrow \infty.$$

In particular, neither steering mechanism can be asymptotically full-state accurate.

Proof Fix $u \neq 0$ and $v \in Y$. Since $\|Q_k\|, \|S_k\| \to 0$, we have $x_k(u, v) \to J_{\text{out}} u$.

The argument uses the same expanding-subspace principle in both cases. We identify a T-invariant subspace $E \subseteq Z$ and a projection $\Pi : Z \to E$ commuting with T such that $\Pi J_{\text{out}} = J_{\text{out}}$ and $\|(T|_E)^{-n}\| \to 0$.

In case (i), take $E = Z$ and $\Pi = I_Z$. Writing $n = ad + r$, with $0 \le r < d$, the cycle identity $W^d = \Omega I_Z$ gives $W^{-n} = \Omega^{-a} W^{-r}$. Consequently, $\|W^{-n}\| \le C|\Omega|^{-a} \to 0$, where $C := \max_{0 \le r < d} \|W^{-r}\|$.

In case (ii), let $E := \bigoplus_{\ell=1}^{p} E_{j_\ell}$ defined as in Appendix A.2 be the direct sum of the selected invariant conjugate planes. We established that the restriction $R|_E$ is similar to $\bigoplus_{\ell=1}^{p} \rho_{j_\ell} \operatorname{Rot}(-\phi_{j_\ell})$. Hence, setting $\rho_{\min} := \min_{1 \le \ell \le p} \rho_{j_\ell} > 1$, there exists $C > 0$ such that $\|(R|_E)^{-n}\| \le C \rho_{\min}^{-n} \to 0$. The projection Π onto E along the remaining invariant spectral subspaces commutes with R. Moreover, the construction of the channels gives $J_{\text{out}} Y \subseteq E$ (by definition of J_{out} and Equations (5) and (6)), so $\Pi J_{\text{out}} = J_{\text{out}}$.

Thus, in either case, $\Pi x_k(u, v) \to J_{\text{out}} u \neq 0$, where the last inequality follows from $P J_{\text{out}} = I_Y$. Since Π commutes with T, we obtain

$$\begin{aligned} \|\Pi\| \, \|T^{n_k} x_k(u, v)\| &\ge \|T^{n_k} \Pi x_k(u, v)\| \\ &\ge \frac{\|\Pi x_k(u, v)\|}{\|(T|_E)^{-n_k}\|} \longrightarrow \infty. \end{aligned}$$

Therefore $\|T^{n_k} x_k(u, v)\| \to \infty$. Finally, the reverse triangle inequality yields

$$\|T^{n_k} x_k(u, v) - J_{\text{out}} v\| \ge \|T^{n_k} x_k(u, v)\| - \|J_{\text{out}} v\| \longrightarrow \infty.$$

This proves the obstruction in both cases. ∎

Appendix C. Proof of Equation (3)

Proposition 18 *For every $c \in Y_{\text{fr}}$ and every $s \ge 0$,*

$$\mathscr{D}_s\left(\mathcal{C}_{\text{scaff}}^{s} c\right) = \mathscr{D}_0(c).$$

Moreover, $\mathcal{C}_{\text{scaff}}^{d_{\text{scaff}}} = \Omega_{\text{scaff}}^{-1} I_{Y_{\text{fr}}}$. Consequently, if $s = a d_{\text{scaff}} + t$, where $0 \le t < d_{\text{scaff}}$, then $\mathcal{C}_{\text{scaff}}^{s} = \Omega_{\text{scaff}}^{-a} \mathcal{C}_{\text{scaff}}^{t}$.

Proof We first establish the one-step identity $\mathscr{D}_1(\mathcal{C}_{\text{scaff}} c) = \mathscr{D}_0(c)$. By definition,

$$\mathscr{D}_1(\mathcal{C}_{\text{scaff}} c) = \sum_{k=1}^{K} \sum_{j=1}^{|D_k|} \sum_{m=0}^{d_{\text{scaff}}-2} c_{k,m+1,j} J_k W_{\text{scaff}}^{m+1} x_{k,j} + \sum_{k=1}^{K} \sum_{j=1}^{|D_k|} \Omega_{\text{scaff}}^{-1} c_{k,0,j} J_k W_{\text{scaff}}^{d_{\text{scaff}}} x_{k,j}.$$

Since $W_{\text{scaff}}^{d_{\text{scaff}}} = \Omega_{\text{scaff}} I_X$, the second sum equals $\sum_{k,j} c_{k,0,j} J_k x_{k,j}$. Relabeling the first sum therefore gives

$$\mathscr{D}_1(\mathcal{C}_{\text{scaff}} c) = \sum_{k=1}^{K} \sum_{m=0}^{d_{\text{scaff}}-1} \sum_{j=1}^{|D_k|} c_{k,m,j} J_k W_{\text{scaff}}^{m} x_{k,j} = \mathscr{D}_0(c).$$

The same calculation, with W_{scaff}^{m+s} in place of W_{scaff}^{m}, yields $\mathscr{D}_{s+1}(\mathcal{C}_{\text{scaff}} c) = \mathscr{D}_s(c)$ for every $s \ge 0$. Iterating this identity gives $\mathscr{D}_s(\mathcal{C}_{\text{scaff}}^{s} c) = \mathscr{D}_0(c)$.

The operator $\mathcal{C}_{\text{scaff}}$ shifts each depth coefficient one position backward in its indexed coefficient cycle, multiplying it by $\Omega_{\text{scaff}}^{-1}$ when it crosses the wrap. After exactly d_{scaff} steps, each coefficient has crossed the wrap once. Hence $\mathcal{C}_{\text{scaff}}^{d_{\text{scaff}}} = \Omega_{\text{scaff}}^{-1} I_{Y_{\text{fr}}}$. Writing $s = a d_{\text{scaff}} + t$ gives the final identity. ∎

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